customer

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Abstract

This repository is a set of custom tools and documentation.

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Customer

Configure BASH for XML API

Export these environment variables to prepare for script execution.

NOTE: You may also specify USER and PAN_IP by passing flags to the script.

```
export USER=<the_name_of_the_firewall_administrator_account_you_configured> # OPTIONAL
export PASS=<the_password_of_the_firewall_administrator_account_you_configured>
export PAN_IP=<the_panorama_instance_ip_address> # OPTIONAL
```

OPTIONAL BASH ENV VAR: USER

This is the username you created previously for the dedicated XML API user.

NOTE: You may also specify a user from the command line with the `-u` flag.

For the `USER` setting, the tool has a default value of `xml-api-user`. To override this value, use the following command before running the tool:

```
export USER="my_new_username" # set the value
echo $USER # validate it was set properly
```

BASH ENVIRONMENT VARIABLE: PASS

This is the password for the new XML API user that was created. It is important that we do not hardcode any active passwords into the script.

For the `PASS` setting, the tool has a default value of `default`. To override this value, use the following command before running the tool:

```
export PASS="my_secret password" # set the value
echo $PASS # validate it was set properly
set | grep $PASS # different way to validate
```

Alternatively, you may choose to use a combination of the direnv and pass tools to make the management of these variables slightly easier.

OPTIONAL BASH ENV VAR: PAN_IP

This is the IPv4 address of the active Panorama.

NOTE:

- You may specify TESTNET Panorama IP from the command line with the `-t` flag.
- You may specify PRODUCTION Panorama IP from the command line with the `-p` flag.

For the `PAN_IP` setting, the tool has a default value of `34.134.31.136`. To override this value, use the following command before running the tool:

```
export PAN_IP="10.2.3.4" # example, set the value
echo $PAN_IP # validate it was set properly
set | grep $PAN_IP # another way to validate
```

Panorama XML API

In order to allow the BASH script to communicate with Panorama via the XML API, you need to enable the API access.

As best practice, set up a separate admin account for the XML API access.

1. Configure an Admin Role Profile:
   - Select `Device` > `Admin Roles` and click `Add`.
   - Enter a `Name` to identify the role.
   - Select the `Role` scope.
   - In the `XML API` tab, click the icon for each functional area to toggle it to the desired setting: `Enable` or `Disable`.
   - Click `OK` to save the profile.
   - Assign the role to an administrator - complete the `Configure a Firewall Administrator Account` step.
2. Configure a Firewall Administrator Account:
   - Select `Device` > `Administrators` and click `Add`.
   - Enter a user `Name` to identify the account. If the firewall uses a local user database to authenticate the account, enter the name that you specified for the account in the database.
   - Select an `Authentication Profile` or sequence if you configured either for the administrator. If the firewall uses `Local Authentication` without a local user database for the account, select `None` (default) and enter a `Password`.
   - Select the `Administrator Type` > `Custom Panorama Admin`.
   - Select the `Admin Role` you created for this account in the `Profile` section.
   - Click `OK` and `Commit`.

NOTE: You must create the admin role and XML user in two places. The Panorama and the FW template/template stack must both have the role and dedicated XML user added.

Validate XML API configuration

To check if the `XML API` has been enabled, run the following code snippet:

```
curl -k -X GET 'https://$PAN_IP//api/?type=keygen&user=$USER&password=$PASS'
```

If everything was configured correctly, a response status `success` should be returned together with an API key for the specified account.

CLI Dev

- Use the CLI to Find XML API Syntax

SCRIPT: `bin/status_panorama.sh`

Disk Status

The script should be updated to check disk status of log collectors.

```
tail follow yes  mp-log vld-1-0.log
tail follow yes  mp-log vld-2-0.log
show system disk-space
less mp-log vldmgr.log
less mp-log logd.log
less mp-log configd.log | match Error
show logging-status all
show system software status | match "elasticsearch\|es-"
show system raid detail
debug log-collector log-collection-stats show incoming-logs
request log-fwd-ctrl device 010108010490 action start-from-lastack
```

SCRIPT: `bin/daily.sh`

- Back Up Configuration and Device State from the CLI

SCRIPT: `bin/deploy_fw_gcp.sh`

- Test without using a public IP. The target environment does not use public IP.
- wait about 10 minutes after script run, login as: `ssh admin@34.67.40.130`

SCRIPT: `bin/stage_release.sh`

The following list can be generated by calling the script with the `-h` flag.

```
franklin@ups:~/bin$ ./stage_release.sh -h

stage_release script.
```

```
Syntax: stage_release.sh [-f|-h|-i|-p|-r|-t|-u|-v]
options:
-f     Specify a file with FW IPv4 addresses, one per line.
-h     Print this Help.
-i     Install release
-p     Use PROD environment
-r     Specify a sofwtware release to download/install (ex: 10.2.0)
-t     Use TESTNET environment
-u     Specify a USER for the Panorama
-v     Enable verbose (debug) mode.
```

Execution

Create a text file and add in IP addresses, one per line. This will allow you to specify exactly which firewalls to operate on.

Example:

```
3.233.53.199
44.216.25.244
34.206.152.182
52.55.185.160
```

To run the script with a list of serials in a file, type the command `./stage_release.sh -f my_ip_list.txt`

To run the script against all firewalls that are managed by this Panorama, type the command `stage_release.sh`

Dev Notes

- Use `pandoc` to generate documentation

```
sudo apt install pandoc
cd docs && pandoc stage_release.md -o stage_release.pdf
```

Jump Host Setup

RHEL/CentOS

```
sudo yum install -y epel-release pandoc git
sudo yum install -y gpg2 # for signing commits, encrypting passwords, etc.
sudo yum install -y direnv # manage env vars
```

pass

the `pass` command as described in this article: https://www.redhat.com/sysadmin/management-password-store

development environment

Run the command: sudo yum group install "Development Tools" Because I would like to configure the development env as detailed here: CentOS / RHEL 7: Install GCC (C and C++ Compiler) and Development Tools - nixCraft (cyberciti.biz)

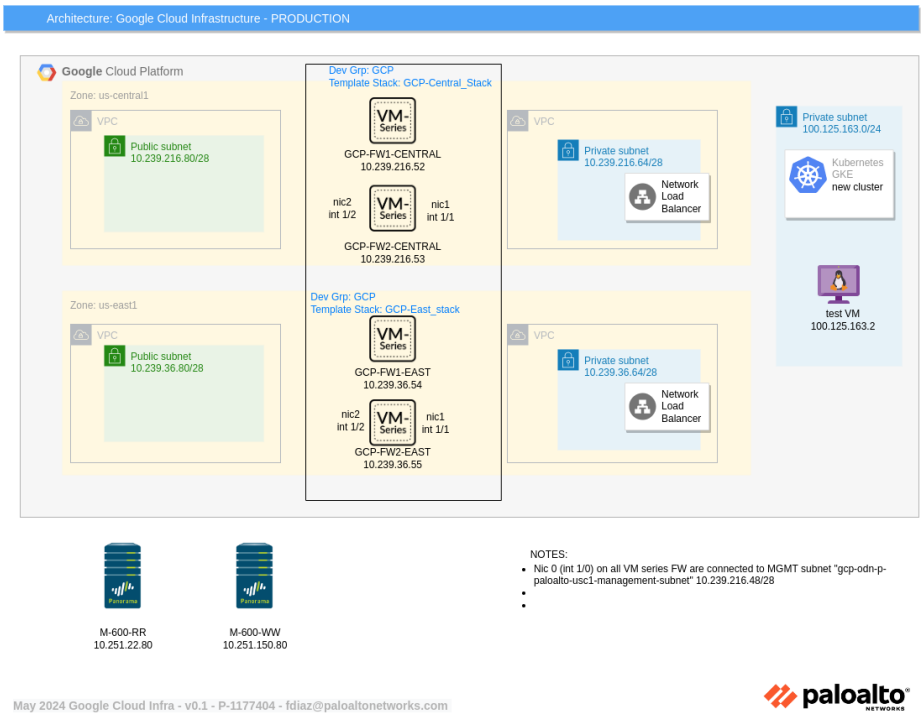
Services

- nginx
- postfix/sendmail
- crontab

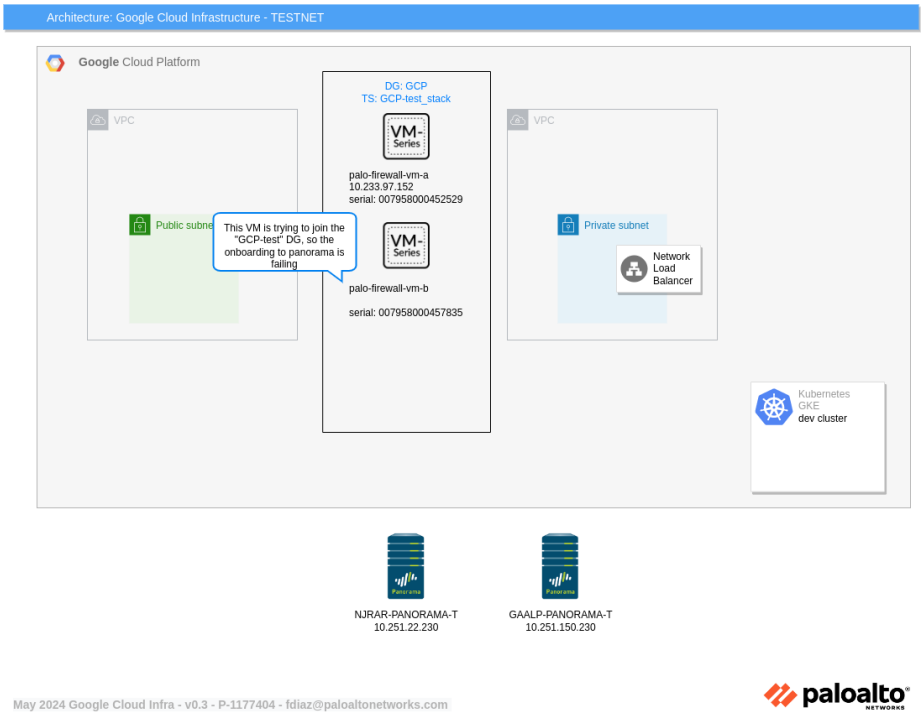
Google Cloud

- Add detail about GCP traffic flows.

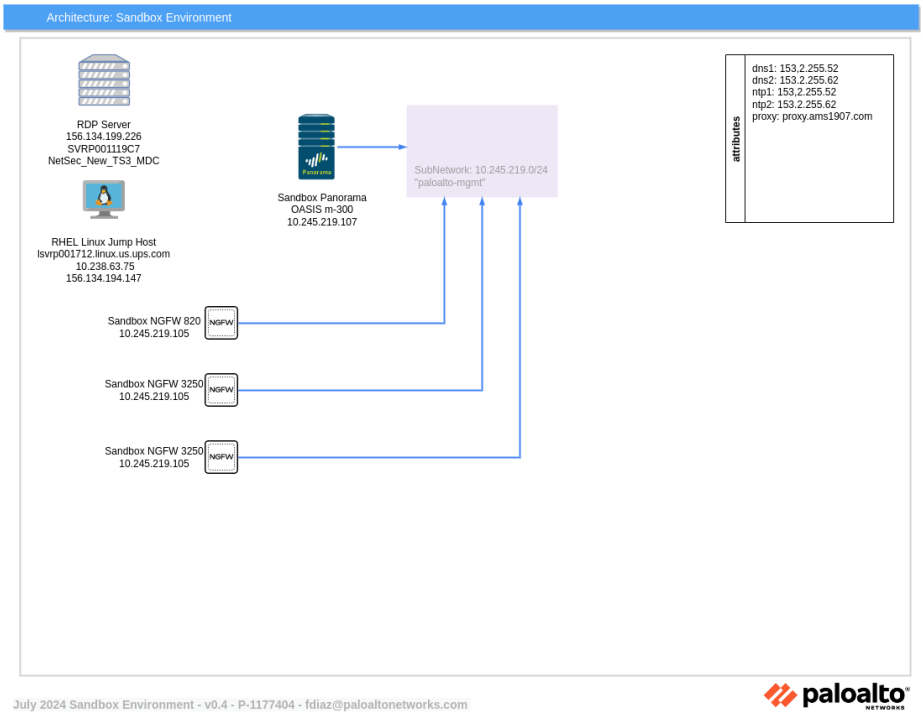
GCP Production



GCP Testnet

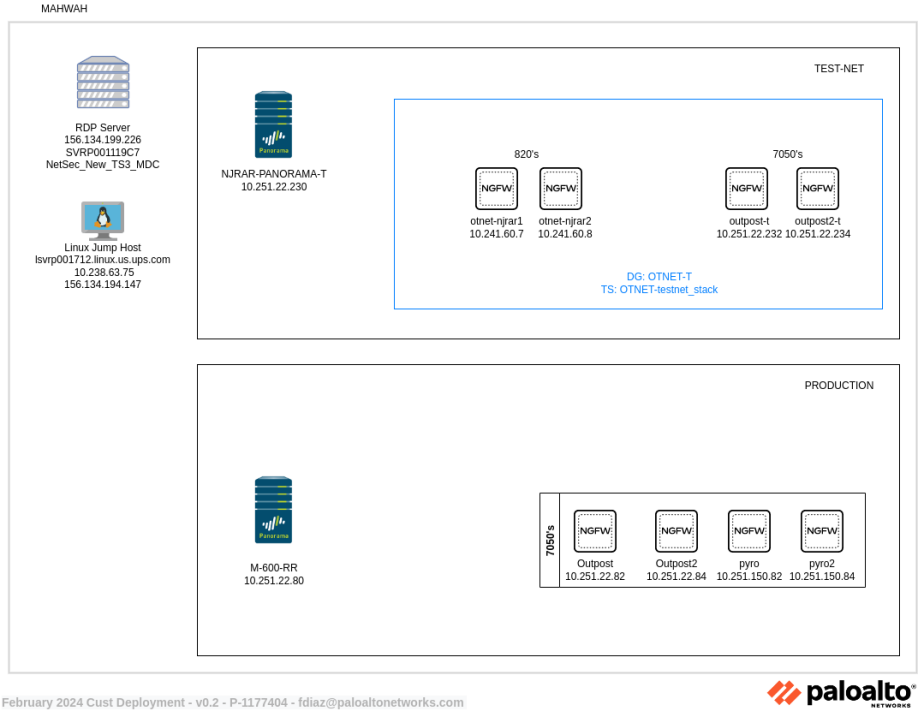

Architecture: Google Cloud Infrastructure - TESTNET
Google Cloud Platform
VPC
Public subne
This VM is trying to join the "GCP-test" DG, so the onboarding to panorama is failing
DG: GCP
TS: GCP-test_stack
VM-Series
palo-firewall-vm-a
10.233.97.152
serial: 007958000452529
VM-Series
palo-firewall-vm-b
serial: 007958000457835
VPC
Private subnet
Network Load Balancer
Kubernetes GKE
dev cluster
NJRAR-PANORAMA-T
10.251.22.230
GAALP-PANORAMA-T
10.251.150.230
May 2024 Google Cloud Infra - v0.3 - P-1177404 - fdiaz@paloaltonetworks.com
paloalto NETWORKS

SSO Testing in GCP



PROD and TESTNET

New Jersey



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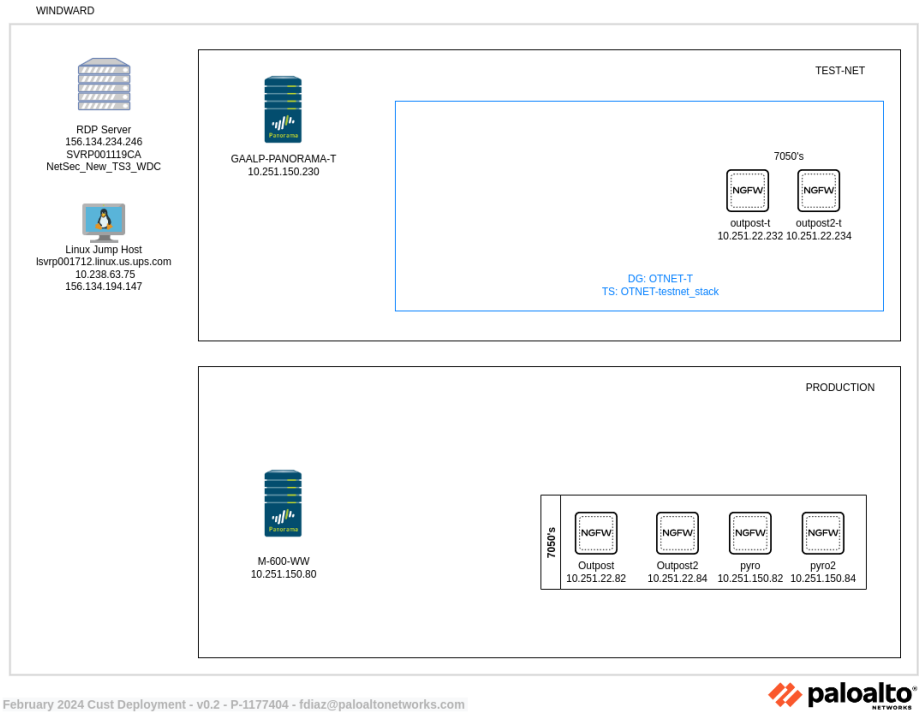
Georgia



Github

Franklin

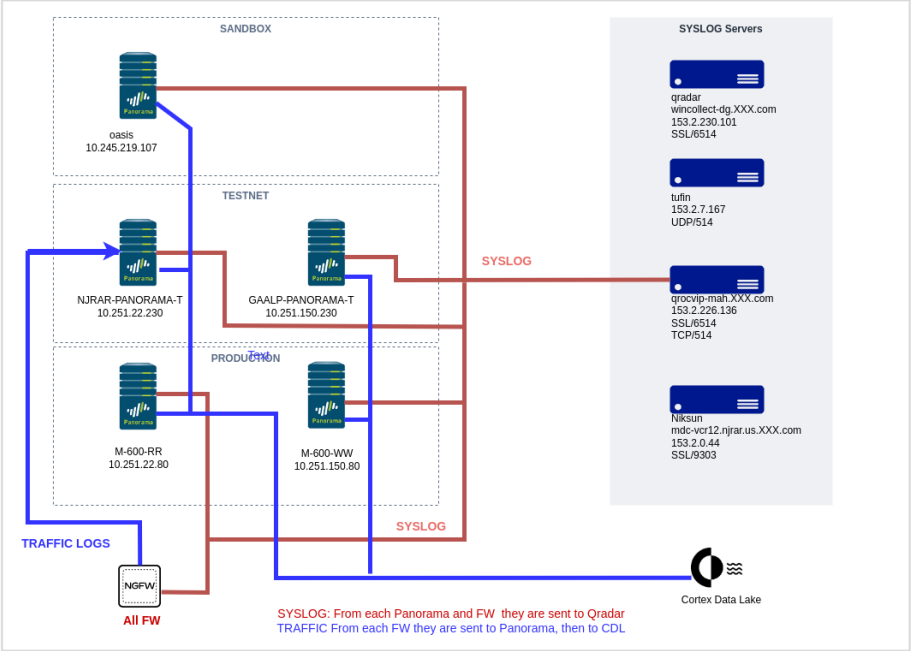
customer internal

Certificates

- An ACME Shell script: acme.sh
- ACME wiki

```
curl https://raw.githubusercontent.com/acmesh-official/acme.sh/master/acme.sh | sh -s -- --i
show device-certificate status
show device-certificate info
```

Logging



QRadar

The QRadar server is at `qrocvip-mah.xxx.com` with IP `153.2.226.136` on TCP port 514.

Syslog

Syslog is configured in two places on the Panorama. Once in the Panorama tab for the Panorama itself, and again in the Device template.

- From the Global Template: `device -> server profiles -> syslog`
  - This makes the firewalls aware of all the SYSLOG servers.
- For SYSLOG coming from the Panorama(s) you should also check `Panorama -> server profiles -> syslog`

Cortex Data Lake

Capture and describe the settings for CDL here.

Troubleshooting

Firewall Log Forwarding

- The log forwaring profiles are managed for each Device Group under `Objects > Log Forwarding`
  - Be sure to select the desired Device Group from the drop down menu.
- Should the "Enable enhanced application logging to Cortex Data Lake" button be checked?

Firewall Syslog Template

- Note that SYSLOG is different from the traffic log forwarding mentioned in the previous section.
- Make sure the IP address of the jump host or other test host is in the permitted mgmt IP address list.
- view details from the firewall using these command below
- You could also try tcpdump from `Panorama > Managed Devices > Troubleshooting`

```
debug syslog-params show
ping host <IP address of syslog server>
traceroute host <IP address of syslog server>
```

- Check the TCP connection between the firewall and the syslog server.
- Replace TCP port if not using TCP/514

```
show netstat numeric-hosts yes numeric-ports yes all yes | match 514
```

- On Panorama Template, check permitted IP addresses: `Device > Setup > Interfaces > Management > Permitted IP Addresses`
  - The IP address for each SYSLOG collector should be listed in the permitted list.